

Zadanie 32

Oblicz granicę $\lim_{n \rightarrow \infty} \frac{\cos(n^2+n)}{5n+1}$.

① Korzystamy z twierdzenia o trzech ciągach.

$$-1 \leq \cos(n^2+n) \leq 1$$

$$\frac{-1}{5n+1} \leq \frac{\cos(n^2+n)}{5n+1} \leq \frac{1}{5n+1}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{5n+1} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{5n+1} = 0$$

$$\text{Zatem } \lim_{n \rightarrow \infty} \frac{\cos(n^2+n)}{5n+1} = 0$$

$$\text{Odp. : } \lim_{n \rightarrow \infty} \frac{\cos(n^2+n)}{5n+1} = 0.$$